

Yihang Jiao

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Research Interests

Machine learning for biomedical data: medical image segmentation, biomedical literature and knowledge retrieval, and reliable evaluation of deep models under class imbalance, limited supervision, and distribution shift.

Education

University of Illinois Urbana-Champaign

Expected: Dec 2027

Master of Computer Science

GPA: 3.91/4.0

University of Illinois Urbana-Champaign

May 2025

B.S. in Mathematics and Computer Science

GPA: 3.83/4.0

Relevant Coursework: Deep Learning, Machine Learning Theory, Applied Machine Learning, Text Mining, Software Engineering for GenAI, Statistics and Probability, Algorithms

Research Experience

FOCAL Lab, UIUC (Prof. Gagandeep Singh)

Aug 2024 – Dec 2024

Research Assistant — Robustness of Neural Language Models

- Built a hybrid adversarial-attack framework pairing LLaMA-2 counterfactual generation with a three-stage semantic filter (label-flip rate ≥ 0.6 , embedding cosine similarity ≥ 0.75 , GPT-2-XL perplexity ≤ 150) to probe classifier robustness beyond word-substitution attacks.
- Filtering discarded 37.7% of generated candidates while retaining 91% of effective adversarial examples, raising attack success from 68.2% to 85.1% on RoBERTa-Large and 71.4% to 79.8% on BERT-Base over TextFooler and PWWS baselines.
- Found that susceptibility scales with model capacity and that inference-style tasks (RTE, QNLI; +11.3% average) gain more from generative rewriting than sentiment tasks, where lexical substitution already suffices; surveyed LoRA backdoors and the cost of stripping safety alignment from open-weight models.

Projects

Brain Tumor Segmentation on Multi-Modal MRI (BraTS): U-Net vs. TransUNet

Aug 2025 – Dec 2025

Independent Project

- Built a unified, reproducible 2D segmentation pipeline over four co-registered MRI modalities (FLAIR, T1, T1-CE, T2) supporting three model families under one entrypoint: a U-Net baseline, a configurable U-Net (depth, width, normalization, residual blocks), and a ResNet50 + ViT TransUNet.
- Designed the biomedical data path end to end: automatic brain-region cropping from non-zero tissue, slice indexing that discards empty slices and flags tumor presence, per-slice z-score intensity normalization, and weighted foreground oversampling to counter severe tumor/background imbalance.
- Isolated architectural effects under fixed data and training settings: redesigning the objective alone (present-class foreground Dice with warmup, class weighting, foreground oversampling) recovered roughly +7 Dice points on an unchanged architecture, showing loss and sampling design matter as much as backbone choice.
- Reached 0.7575 foreground Dice with TransUNet vs. 0.7326 for the tuned U-Net, trained from scratch under a 15-epoch budget with mixed precision and gradient accumulation; class-wise analysis localized the transformer's advantage to the diffuse tumor subregion (Dice 0.7652 vs. 0.6800) rather than to compact structures.

VibeRepair+: Retrieval-Augmented Program Repair for Java and Python

Jan 2026 – May 2026

CS Software Engineering for GenAI — Co-author

- Designed a pluggable retriever framework over a 97,402-example bug-fix corpus in which adding a retrieval strategy requires one Python file and zero changes to the serving endpoint, evaluation harness, or metrics; reproducibility enforced by a corpus SHA-256 fingerprint and locked holdout split.
- Diagnosed the incumbent leave-one-out evaluation as a self-match artifact (perfect scores despite only 34.1% top-1 agreement and 0.166 top-10 Jaccard between retrievers) and replaced it with an external-query protocol on 521 real bugs, using an operator-aware token-overlap proxy ground truth and a two-stage embedding + edit-distance leak check.
- Under the corrected protocol, lexical BM25 beat the dense-embedding baseline by 27.8% relative recall@10, inverting the usual dense-beats-lexical assumption; a Reciprocal Rank Fusion hybrid ($k = 60$) led five of six quality metrics and recovered 15 top-10 hits missed by both sub-retrievers.

LLM RAG Reranker for Scientific Literature with Citation Context

Aug 2024 – Dec 2024

CS 598 Text Mining — Co-author

- Built a literature retrieval system that reranks candidate papers via LLM-adjudicated pairwise comparisons, treating noisy LLM judgments as a dueling-bandit problem and applying UCB Elimination to cut comparisons from $O(N^2)$ (190 exhaustive pairs at $K=20$) to roughly 44.
- Curated a corpus of 10,395 papers and 497,210 citation contexts via breadth-first traversal of LaTeX sources; augmenting candidates with in-link citation context raised MRR from 0.485 to 0.572 at $K=5$, outperforming reranking on titles and abstracts alone.
- Built a retrieval-augmented writing pipeline reaching 0.64 SummaConv factual consistency against source material, close to the 0.65 scored by human-written text, and 0.87 semantic similarity to the human reference.

Skills

Machine Learning: PyTorch, CUDA, HuggingFace Transformers, mixed-precision and multi-GPU training, gradient accumulation, 4-bit quantization (BitsAndBytes), FAISS, scikit-learn, NumPy, pandas

Biomedical & Scientific Data: multi-modal MRI volume handling and intensity normalization, 2D/3D semantic segmentation, Dice-based evaluation under class imbalance, large-scale scientific corpus construction

Programming: Python, C++, C#, Java, Git, $\LaTeX$